

Day 4: Open Bank Account

1. Day Number

Day 4

2. Topic Name

Data Types and Type Conversion

Today you will learn how Python stores different kinds of information and how to convert user input into numbers.

3. Connection with the Previous Day

Previously, you checked whether an entered password matched the saved password.

After a successful password login, you will now create a simple bank account by collecting:

- Account holder's name
- Account number
- Initial balance

4. Important Topics

- String
 - Integer and decimal numbers
 - `input()`
 - `int()`
 - `float()`
 - `print()`
 - Type conversion
 - Functions
-

5. Foundational Notes

What is a data type?

A **data type** tells Python what kind of value is stored in a variable.

```
name = "Rahul"  
age = 35  
balance = 1500.50
```

Here:

- "Rahul" is a **string**
- 35 is an **integer**
- 1500.50 is a **float**

String

A string stores text.

```
account_holder = "Rahul Tiwari"
```

Names can contain spaces, so they should be stored as strings.

Integer

An integer stores a whole number without a decimal point.

```
number_of_accounts = 2
```

Float

A float stores a number that may contain a decimal point.

```
balance = 2500.75
```

A bank balance should normally be converted to float because it may include cents, such as \$500.50.

Important behaviour of input()

The input() function always returns a **string**, even when the user enters a number.

```
balance = input("Enter balance: ")
```

If the user enters 500.50, Python initially stores it as:

```
"500.50"
```

This is text, not a numeric value.

Type conversion

Type conversion changes a value from one data type to another.

Convert text into an integer:

```
age = int("35")
```

Convert text into a decimal number:

```
price = float("99.50")
```

Why keep an account number as a string?

Although an account number contains digits, you usually do not perform mathematical calculations on it.

It may also begin with zero:

```
0012345678
```

Converting it to an integer could remove the starting zeros. Therefore, the account number should generally remain a string.

6. Easy Example

The following example demonstrates type conversion without solving today's complete problem:

```
def display_product():
    product_name = input("Enter product name: ")
    price_text = input("Enter product price: ")

    price = float(price_text)

    print("Product:", product_name)
    print("Price:", price)
```

What happens here?

1. The program collects the product name as a string.
 2. It collects the price as a string.
 3. `float()` converts the price into a decimal number.
 4. The program displays both values.
-

7. Problem Statement

Create a Python program that:

1. Asks the user to enter the account holder's name.
2. Asks the user to enter the account number.
3. Asks the user to enter the initial balance.
4. Converts the initial balance into a number.
5. Displays the complete account details.

Use a **functional approach** by placing the main logic inside a function.

8. Concepts Used

Variables

Variables store the entered account information.

Possible variables:

```
account_holder_name
account_number
initial_balance
```

Strings

The following values should be strings:

- Account holder name
- Account number

Float

The initial balance can be stored as a float because it may contain a decimal value.

Input

Use `input()` to collect information from the user.

Output

Use `print()` to display the created account details.

Type conversion

Use `float()` to convert the balance from a string into a number.

Function

Put the account-creation steps inside a function, such as:

`create_bank_account`

9. Thought Process

Step 1: Create a function

Create one function responsible for opening the bank account.

Step 2: Ask for the account holder's name

Store the complete name as a string. The name may contain spaces.

Step 3: Ask for the account number

Keep the account number as a string because it is an identifier, not a value used in calculations.

Step 4: Ask for the initial balance

Remember that `input()` returns the balance as a string.

Step 5: Convert the balance

Use `float()` to convert the entered balance into a numeric value.

For example:

`"1500.75" → 1500.75`

Step 6: Display the account details

Print:

- Account holder name
- Account number
- Initial balance

Step 7: Call the function

After defining the function, call it so that the program runs.

10. Pseudocode

```
START

DEFINE a function to create a bank account

    ASK the user for the account holder name
    STORE it as a string

    ASK the user for the account number
    STORE it as a string

    ASK the user for the initial balance
    CONVERT the balance to a float

    DISPLAY "Account Created Successfully"
    DISPLAY the account holder name
    DISPLAY the account number
    DISPLAY the initial balance

CALL the function

END
```

11. Suggested Solving Approach

Use a **functional approach**.

Recommended structure:

```
Define create_bank_account function
    Collect all inputs
    Convert the balance
    Display the details

Call create_bank_account function
```

For this beginner exercise, one function is enough.

12. Easy Edge Cases

Balance is 0

Input:

0

This should be valid because a bank account may be opened with a zero initial balance.

After conversion:

0.0

Name contains spaces

Input:

```
Rahul Kumar Tiwari
```

`input()` accepts the complete line, including spaces, so the full name can be stored correctly.

Account number begins with zero

Input:

```
0012345678
```

Keep it as a string so that the starting zeros are preserved.

Decimal balance

Input:

```
1500.75
```

Use `float()` so that the decimal part is retained.

13. Expected Input

```
Enter account holder name: Rahul Kumar  
Enter account number: 0012345678  
Enter initial balance: 1500.50
```

14. Expected Output

```
Account Created Successfully  
  
Account Holder: Rahul Kumar  
Account Number: 0012345678  
Initial Balance: $1500.50
```

The exact spacing or formatting may differ, but all three account details should be displayed.

15. Hint Only

Create a function and collect three values using `input()`.

Remember:

```
name → keep as string  
account number → keep as string  
balance → convert using float()
```

After defining the function, do not forget to call it.